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(54) INTEGRATED CIRCUIT STRUCTURE AND METHOD FOR MANUFACTURING THE SAME

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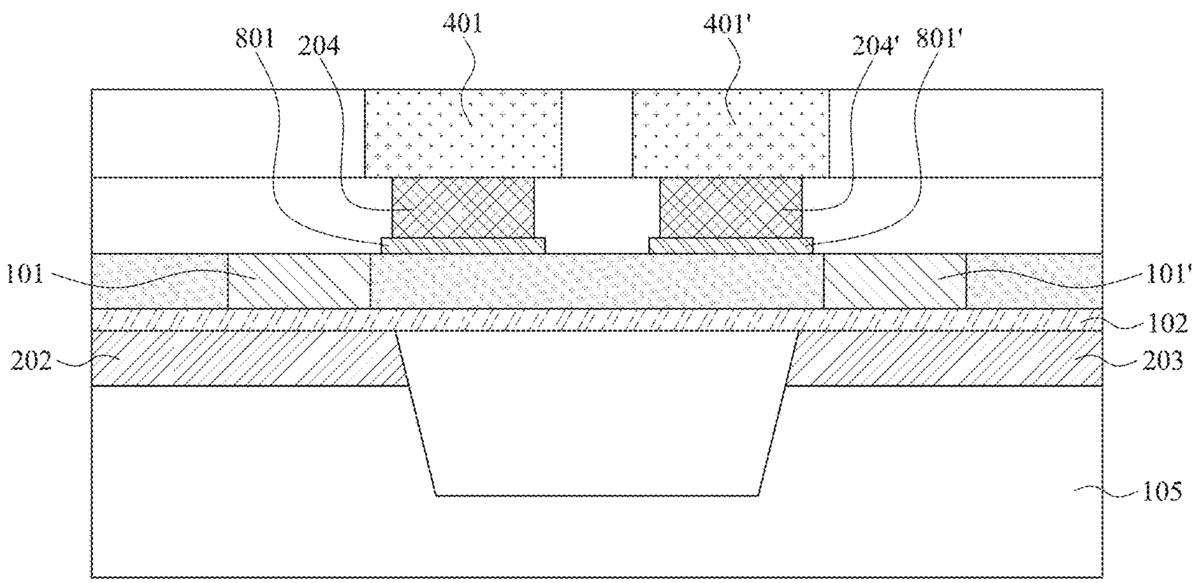
(57) ABSTRACT

An integrated circuit structure includes a first transistor, a second transistor, a first via and a second via. The first transistor includes a first active region and a first gate electrode overlapping the first active region. The second transistor includes a second active region and a second gate electrode overlapping the second active region. The first via is disposed over the first transistor and the first gate electrode. The second via is disposed over the second transistor and the second gate electrode. The first gate electrode and the second gate electrode are arranged mirror-symmetrically with respect to a symmetry plane.



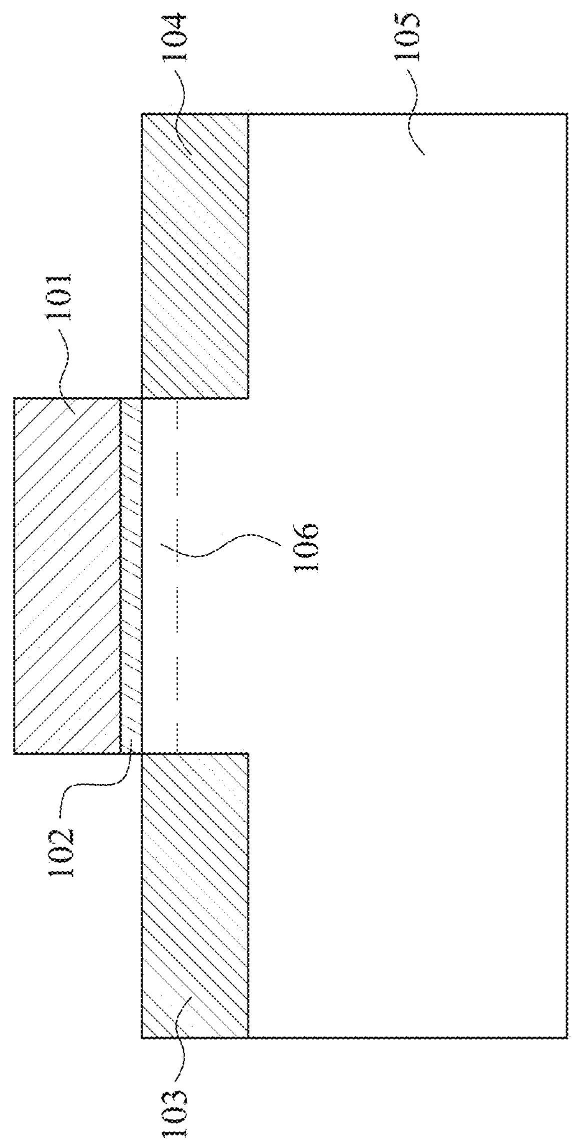


FIG. 1

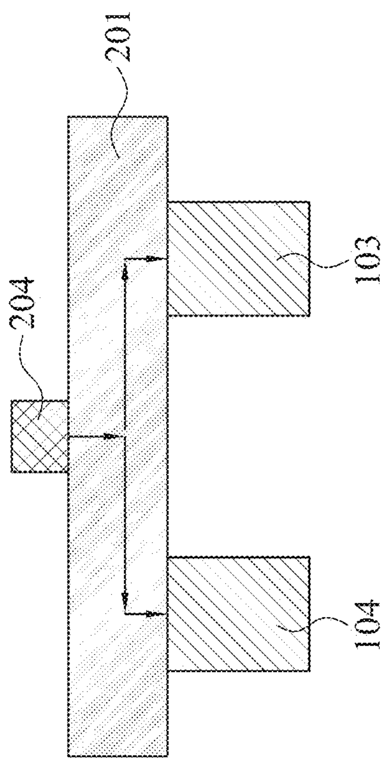


FIG. 2B

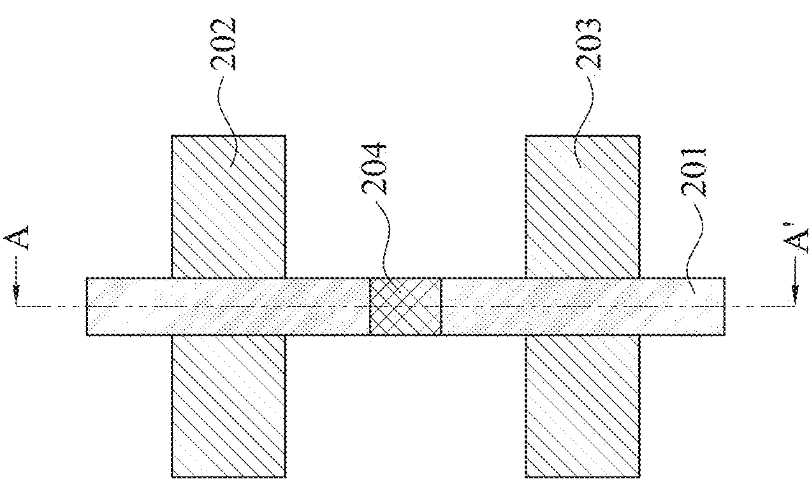


FIG. 2A

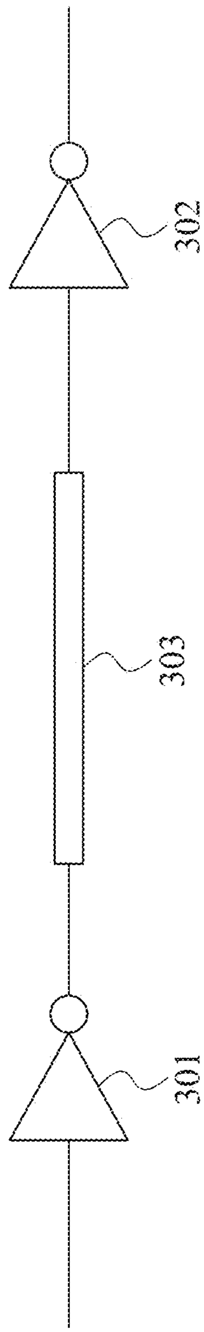


FIG. 3A

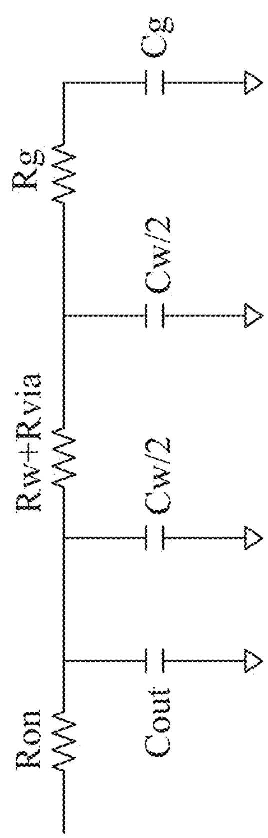


FIG. 3B

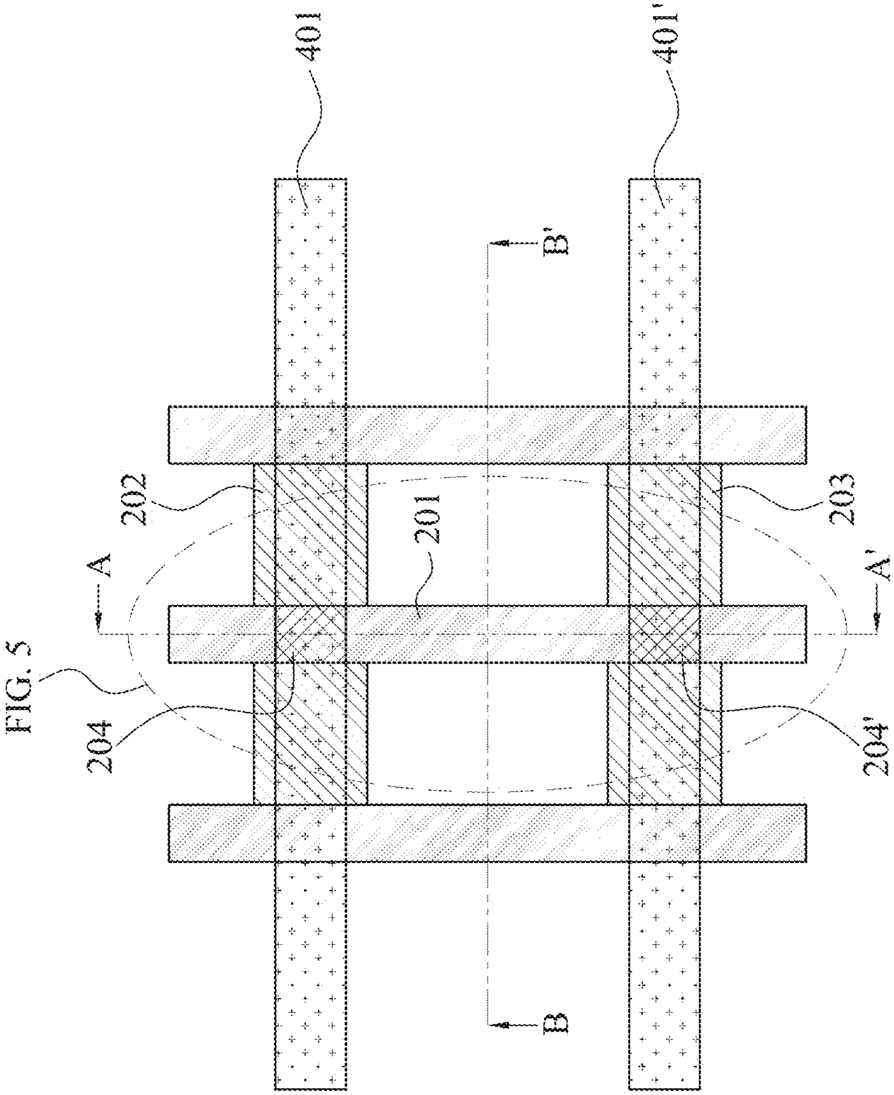


FIG. 4

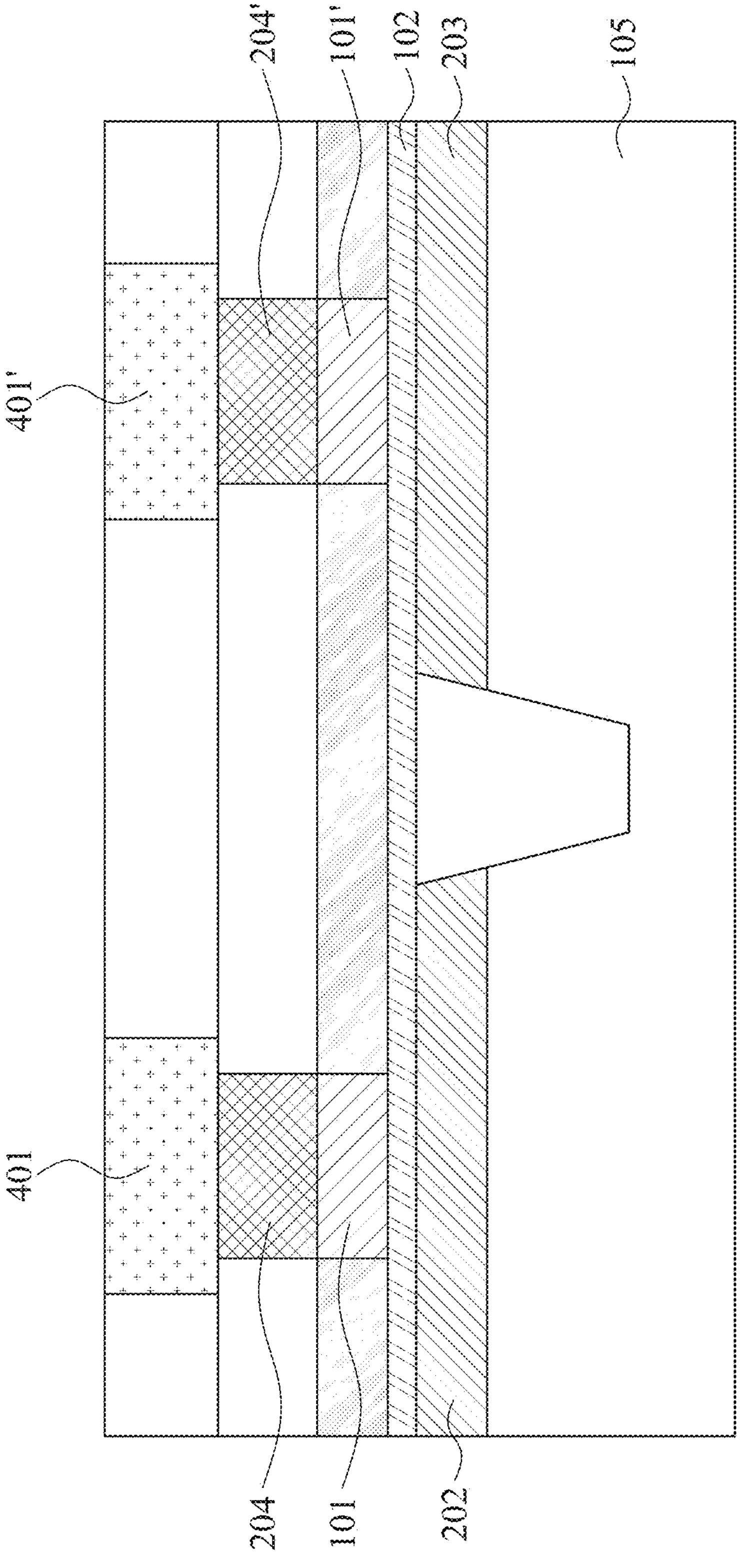


FIG. 5

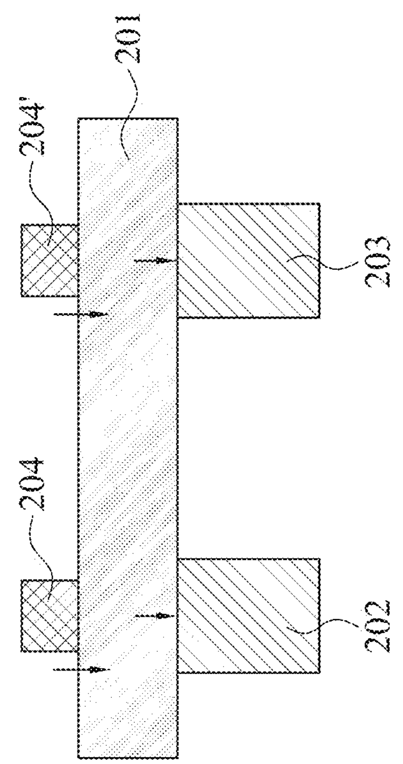


FIG. 6B

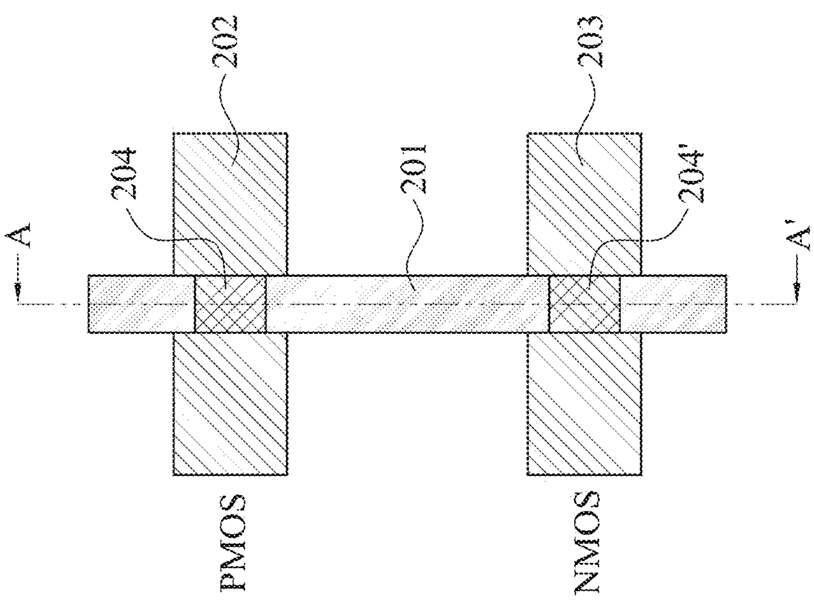


FIG. 6A

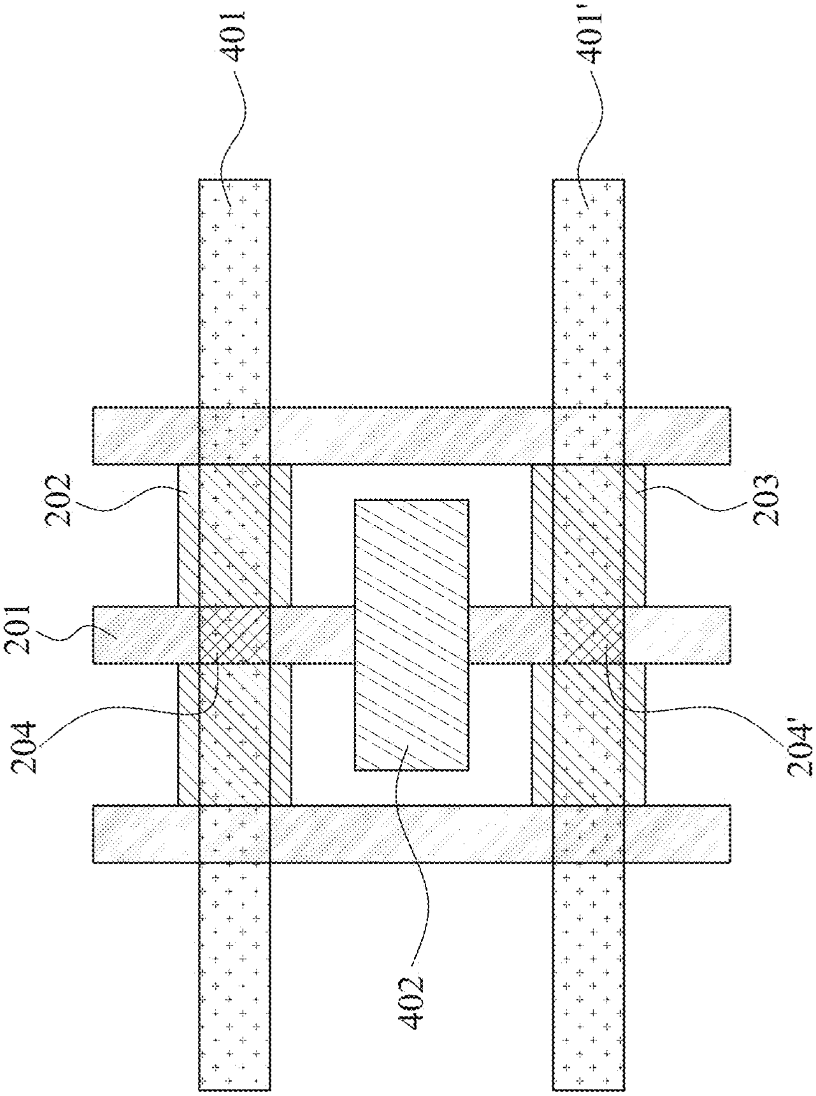


FIG. 7

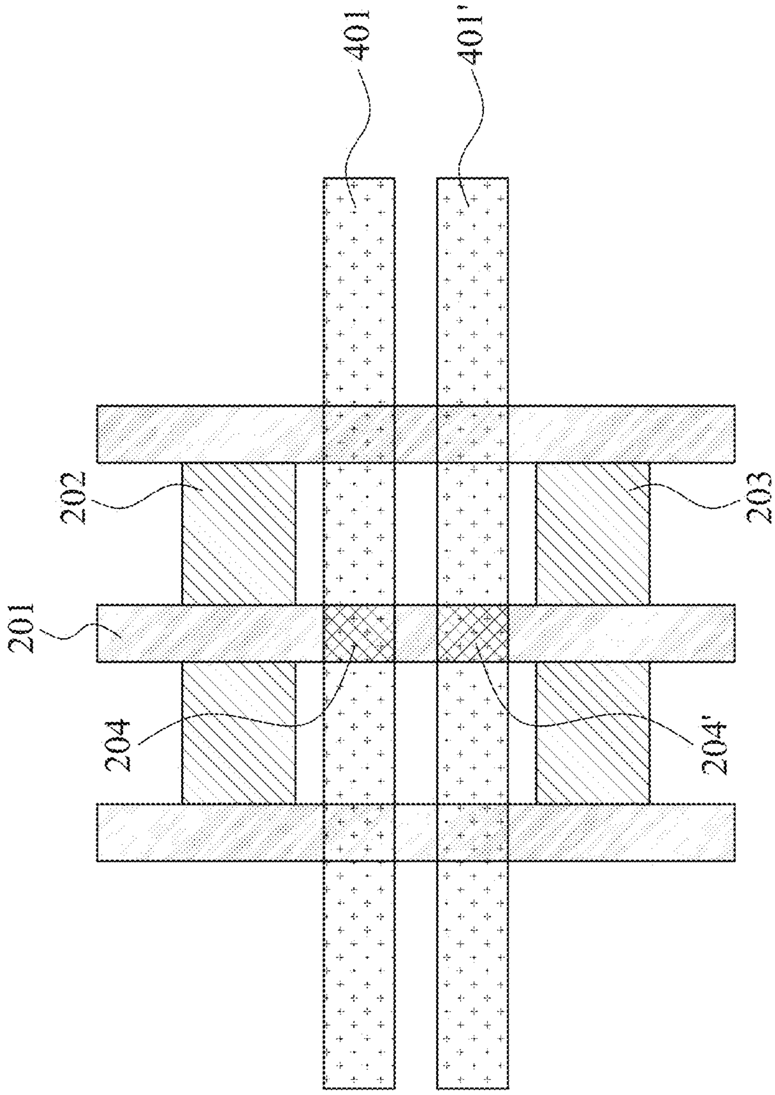


FIG. 8A

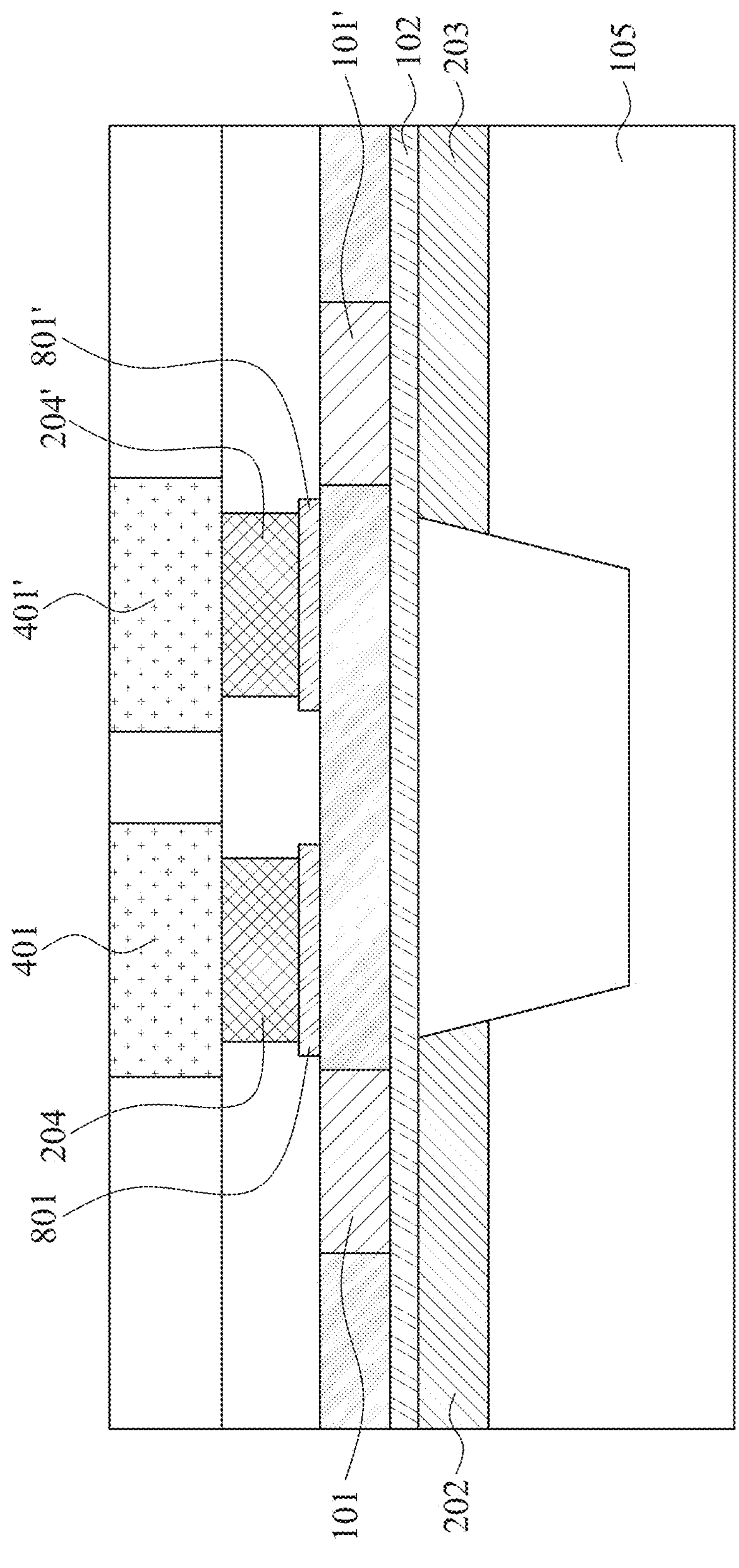


FIG. 8B

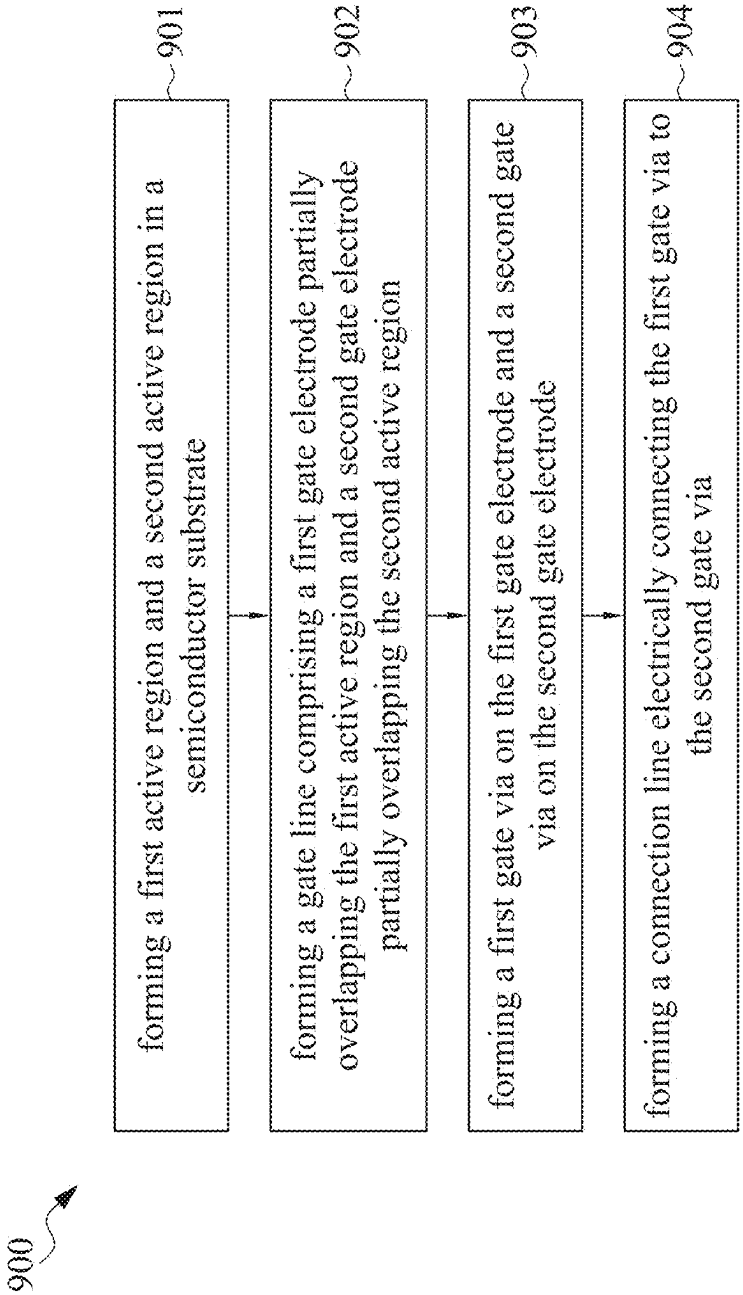

900
901
forming a first active region and a second active region in a semiconductor substrate
902
forming a gate line comprising a first gate electrode partially overlapping the first active region and a second gate electrode partially overlapping the second active region
903
forming a first gate via on the first gate electrode and a second gate via on the second gate electrode
904
forming a connection line electrically connecting the first gate via to the second gate via

FIG. 9

INTEGRATED CIRCUIT STRUCTURE AND METHOD FOR MANUFACTURING THE SAME

PRIORITY DATA

[0001] The present application is a divisional of U.S. patent application Ser. No. 18/312,586 filed on May 4, 2023, now U.S. Pat. No. 12,315,861, issued May 27, 2025, which is a continuation of U.S. patent application Ser. No. 17/037,447 filed on Sep. 29, 2020, now U.S. Pat. No. 11,682,671, issued Jun. 20, 2023, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/928,776 filed on Oct. 31, 2019, the disclosures of which are incorporated herein by reference in their entireties.

BACKGROUND

[0002] Integrated circuits, which comprise a large number of transistors, specifically, metal-oxide-semiconductor ("MOS") transistors, are widely used in electronic devices such as cell phones, computers and digital home appliances. Growing demand for smaller and thinner electronic devices with more functions leads to the research and development of transistors with smaller dimensions in order to allow more transistors to be packed within a device in a greater amount to perform more functions.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. **1** shows a schematic MOS transistor structure in accordance with some embodiments of the present disclosure.

[0005] FIG. **2**A shows exemplary plan view of an integrated circuit layout in accordance with some embodiments of the present disclosure.

[0006] FIG. **2**B shows a cross-sectional view of the integrated circuit structure of FIG. **2**A.

[0007] FIG. **3**A shows a schematic integrated circuit structure in accordance with some embodiments of the present disclosure.

[0008] FIG. **3**B shows the circuit diagram of the schematic integrated circuit structure of FIG. **3**A.

[0009] FIG. **4** shows exemplary plan view of the aforementioned integrated circuit structure in accordance with some embodiments of the present disclosure.

[0010] FIG. **5** shows the cross-sectional view of the integrated circuit structure of FIG. **4**.

[0011] FIG. **6**A shows a partial and simplified enlargement of the integrated circuit structure shown in FIGS. **4** and **5**.

[0012] FIG. **6**B shows a cross-sectional view of the structure shown in FIG. **6**A.

[0013] FIG. **7** also shows exemplary plan view of an integrated circuit structure in accordance with some embodiments of the present disclosure.

[0014] FIG. **8**A shows exemplary plan view of an integrated circuit structure in accordance with some embodiments of the present disclosure.

[0015] FIG. **8**B shows the cross-sectional view of the integrated circuit structure of FIG. **8**A.

[0016] FIG. **9** is a flow chart of a method for manufacturing an integrated circuit structure in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0017] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0018] Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0019] As used herein, the terms "approximately," "substantially," "substantial" and "about" are used to describe and account for small variations. When used in conjunction with an event or circumstance, the terms can refer to instances in which the event or circumstance occurs precisely as well as instances in which the event or circumstance occurs to a close approximation. For example, when used in conjunction with a numerical value, the terms can refer to a range of variation of less than or equal to ±10% of that numerical value, such as less than or equal to ±5%, less than or equal to ±4%, less than or equal to ±3%, less than or equal to ±2%, less than or equal to ±1%, less than or equal to ±0.5%, less than or equal to ±0.1%, or less than or equal to ±0.05%. For example, two numerical values can be deemed to be "substantially" the same or equal if a difference between the values is less than or equal to ±10% of an average of the values, such as less than or equal to ±5%, less than or equal to ±4%, less than or equal to ±3%, less than or equal to ±2%, less than or equal to ±1%, less than or equal to ±0.5%, less than or equal to ±0.1%, or less than or equal to ±0.05%. For example, "substantially" parallel can refer to a range of angular variation relative to 0° that is less than or equal to ±10°, such as less than or equal to ±5°, less than or equal to ±4°, less than or equal to ±3°, less than or equal to ±2°, less than or equal to ±1°, less than or equal to ±0.5°, less than or equal to ±0.1°, or less than or equal to ±0.05°. For example, "substantially" perpendicular can refer to a range of angular variation relative to 90° that is less than or equal to ±10°, such as less than or equal to ±5°, less than or equal to ±4°, less than or equal to ±3°, less than or equal to ±2°, less than or equal to ±1°, less than or equal to ±0.5°, less than or equal to ±0.1°, or less than or equal to ±0.05°.

[0020] Referring to FIG. **1**, FIG. **1** schematically shows a MOS transistor structure including a gate electrode **101**, a gate dielectric layer (or "gate oxide layer") **102**, and a source region **103**, and a drain region **104** formed in a substrate **105**. In some embodiments, the source region **103** and drain region **104** are formed in the substrate **105** by doping impurities into the substrate **105** to define the source region **103** and the drain region **104**. The gate dielectric layer **102** located under the gate electrode **101** acts as an insulator to prevent electrons transfer from the gate electrode **101** to the substrate **105**, or from the source region **103**/drain region **104** to the gate electrode **101**, in accordance with some embodiments of the present disclosure. In some embodiments, the gate dielectric layer **102** may be formed by thermal oxidation of the substrate **105** to form a thin layer of gate dielectric layer.

[0021] Types of the MOS transistors may include: a) positive-channel MOS transistor ("PMOS") and b) negative-channel MOS transistor ("NMOS"). A combination of a PMOS transistor and an NMOS transistor appearing in pair and being used in a complementary way to form an effective means of electrical control is called a complementary metal-oxide-semiconductor ("CMOS") transistor. For a PMOS transistor, the substrate **105** is an n-doped silicon substrate with a p-doped source region **103** and a p-doped drain region **104**, respectively, and for an NMOS transistor, the substrate **105** is a p-doped substrate with an n-doped source region **103** and an n-doped drain region **104**, respectively.

[0022] The MOS transistor is operated by applying voltage to the MOS transistor to control the "on" and "off" of the MOS transistor for transmitting/interrupting signals. In some embodiments, the MOS transistor shown in FIG. **1** is an NMOS transistor, when the applied gate voltage is below the threshold for making a conductive channel below the gate electrode **101** and gate dielectric layer **102**, there is little or no conduction between the source region **103** and drain region **104**; the switch of the NMOS transistor is off. However, when the applied gate voltage is over the threshold, the gate attracts electrons, inducing an n-type conductive channel **106** in the substrate below the gate dielectric **102**, which allows electrons to flow between the source region **103** and the drain region **104** and the switch is on.

[0023] In some embodiments, the MOS transistor shown in FIG. **1** is a PMOS transistor, when the applied gate voltage is below the threshold for making a conductive channel below the gate electrode **101** and gate dielectric layer **102**, there is little or no conduction between the source region **103** and drain region **104**; the switch of the PMOS transistor is off. However, when the applied gate voltage is over the threshold, the gate attracts electrons, inducing a p-type conductive channel **106** in the substrate below the gate dielectric **102**, which allows electrons to flow between the source region **103** and the drain region **104** and the switch is on.

[0024] In some embodiments, the material of the gate **101** is a metal, such as tantalum, tungsten, tantalum nitride, and titanium nitride, or polycrystalline silicon ("polysilicon," or "POLY") and the substrate is silicon. In some embodiments where the material of the gate **101** is a tantalum, tungsten, tantalum nitride, and titanium nitride, the material of the gate dielectric is a high-k dielectric. For a MOS transistor, a connection line ("metal wires"; "interconnect(s)") is used for electrically connecting two or more transistors together. Specifically, the connection line and the transistors are electrically connected by a conductive via, or "via contact" or "gate via" located between the gate electrode and the connection line.

[0025] The scaling-down of the dimensions of the transistors basically follows the Moore's Law, which describes the empirical regularity that the number of transistors on integrated circuit doubles approximately every two years. In last few decades, the development of very-large-scale integration ("VLSI") or ultra-large-scale integration ("ULSI"), i.e., the method of making an integrated circuit by combining huge numbers (such as millions or even billions) of transistors, specifically, MOS transistors, onto a single chip, enables complex semiconductor and telecommunication technologies to be developed.

[0026] However, as the gate length and gate height of the transistors are scaling down to meet the requirements for electronic devices of smaller sizes with more functions, it is found that the gate resistance is scaling up and the RC delay increases. The RC delay, or "RC time constant ("τ"), refers to the time required to charge the capacitor, through the resistor, from an initial charge voltage of zero to approximately 63.2% of the value of an applied DC voltage, or to discharge the capacitor through the same resistor to approximately 36.8% of its initial charge voltage, which is equal to the product of the circuit resistance and the circuit capacitance. The RC time constant can be described in the following formula: $\tau=R*C$, where R refers to the total resistance of the integrated circuit and the unit of R is ohm; C refers to the total capacitance, and the unit of C is farads, and the unit of the product of R*C, τ, is seconds.

[0027] Referring to FIGS. **2**A and **2**B, FIG. **2**A shows exemplary plan view of an integrated circuit layout including two transistors electrically connected to the connection line (not shown in the figure) through a conductive via (or "contact via") in accordance with some embodiments of the present disclosure, and FIG. **2**B shows a cross-sectional view of the integrated circuit structure as shown in FIG. **2**A along the AA' line. In FIG. **2**A, a gate line **201** extending in a first direction is arranged above the active regions **202** and **203**, wherein the active regions **202** and **203** both extend in a second direction which is orthogonal to the first direction, and a conductive via **204** is located on the gate line **201**. The gate electrodes (not shown) are formed along the gate line within the area overlapping the active regions **202** and **203**, and the source regions and drain regions for each of the two transistors are formed on the active regions **202** and **203** within the areas adjacent to the area covered by the gate line **201** by doping n-type or p-type impurities to define the source/drain regions as needed. A conductive via **204** is formed on the gate line **201** to electrically connect the gate electrodes to the connection line (not shown) formed above the transistors for applying voltage(s) to the transistors to electrically control the switch on/off of the transistors.

[0028] In some embodiments, the integrated circuit structure shown in FIG. **2**A is a CMOS transistor including both a PMOS transistor (not shown in FIG. **2**A) on the active region **202** and an NMOS transistor (not shown in FIG. **2**A) on the active region **203**, wherein the PMOS transistor and the NMOS transistor are electrically connected to a connection line (not shown) by the single conductive via **204** located on the surface of the gate line **201** within the area between the position of the gate electrodes of the PMOS transistor and the NMOS transistor. In some embodiments, the active regions **202** and **203** include oxide-diffused regions. In some embodiments, the oxide-diffused regions refer to regions that are oxide-based which are further diffused by n-type or p-type impurities to form the drain/source regions of the PMOS and/or NMOS transistors.

[0029] Referring to FIG. **2**B, the arrows in FIG. **2**B refer to the current flow from the conductive via **204** of the integrated circuit structure of FIG. **2**A to the gate electrodes (not shown) which leads to the creation of a channel under the gate electrodes for carriers (electrons/holes) to flow from the source regions to the drain regions when a voltage is applied to the transistors to switch on the transistor(s). As the current flows, resistances are measured, which include a conductive via resistance, sheet resistances along the gate line **201**, and gate resistances of both transistors.

[0030] In advanced semiconductor manufacturing operations, it is found that the impact of the gate resistance in the overall RC delay performance increases as the integrated circuits are down-scaled. Referring to FIGS. **3**A and **3**B, which are to demonstrate the delay calculation of the gate and the interconnect attached to it. FIG. **3**A shows a schematic integrated circuit structure of two inverters **301** and **302** electrically connected by the interconnect **303** in accordance with some embodiments of the present disclosure. In some embodiments, the inverters are CMOS inverters. FIG. **3**B shows the circuit diagram of the schematic integrated circuit structure of FIG. **3**A, wherein the RC delay of the structure is calculated by the following formula: $\tau=R_{on}*(C_{out}+C_g+C_w)+(R_w+R_{via})*(C_g+0.5C_w)+R_g*C_g$, where R_{on} refers to transistor resistance, R_w refers to wire resistance, R_{via} refers to via resistance, C_g refers to input pin capacitance, and R_g refers to gate input resistance. It is found that from the 7 nm semiconductor technology node to the down-scaled 5 nm semiconductor technology node, the gate resistance impact increases more than twofold in the overall RC delay performance. Specifically, gate resistance has more impact in high track library because of the longer gate line which contributes to higher gate capacitance.

[0031] In some embodiments of the present disclosure, an integrated circuit structure including two transistors, such as a CMOS transistor including a pair of a PMOS transistor and an NMOS transistor, two conductive vias electrically connecting to the two transistors, respectively, and a connection line electrically connecting the two conductive vias, is provided. Referring to FIG. **4** and FIG. **5**, FIG. **4** shows exemplary plan view of the aforementioned integrated circuit structure having two transistors with two conductive vias and a connection line and FIG. **5** shows the cross-sectional view of the integrated circuit structure of FIG. **4** along the AA' line. The integrated circuit structure includes a first transistor (not shown in the figure) and a second transistor (not shown in the figure), wherein the first transistor includes a first active region **202**, a first gate electrode **101** over the first active region **202**; and a first channel (not shown) in the first active region **202** and under the first gate electrode **101**, wherein the second transistor includes a second active region **203**, a second gate electrode **101'** over the second active region **203**; and a second channel (not shown) in the second active region and under the second gate electrode; a first conductive via **204** electrically connected to the first gate electrode; a second conductive via **204'** electrically connected to the second gate electrode; and a connection line including but not limited to connection wires **401** and **401'** electrically connecting the first conductive via **204** and the second conductive via **204'**, and wherein the first transistor and the first conductive via **204** and the second transistor and the second conductive via **204'** are arranged mirror-symmetrically with respect to a symmetry plane along line BB' as shown in FIG. **4**. In the following paragraphs, the description "the connection line **401** and **401'**" is simply used as an abbreviation of the description "a connection line including but not limited to connection wires **401** and **401'**, and does not make any restriction to the total numbers of the connection wires and vias electrically connecting the connection wires the connection line includes.

[0032] The active regions **202** and **203** are the regions on the substrate (e.g. silicon) for forming source/drain regions. In some embodiments, the first transistor further includes a pair of first source/drain regions in the first active region **202**, wherein the source region and the drain region are formed on the active region on opposite sides of the first gate electrode, as the arrangement of the source region **103** and the drain region **104** shown in FIG. **1**. Similarly, in some embodiments, the second transistor further includes a pair of second source/drain regions in the second active region **203** and on opposite sides of the second gate electrode.

[0033] Referring back to FIG. **4**, the conductive vias **204** and **204'**, which are designed to connect the gate electrodes of the transistors to the connection line **401** and **401'**, are formed on the gate line **201** to electrically connect the gate electrodes. To control the pair of transistors of the integrated circuit structure of FIG. **4**, the conductive vias are formed on the surface of the gate line **201** within the areas of the surface of the gate electrodes and the area between the two gate electrodes. In some embodiments, the first conductive via **204** is formed on the gate line **201** overlapping the first gate electrode and the first channel, and the second conductive via **204'** is formed on the gate line **201** overlapping the second gate electrode and the second channel.

[0034] In some embodiments, the first conductive via **204** is located on the first gate electrode within an area having a vertical projection apart from the first active region and the first channel, and the second conductive via **204'** is located on the second gate electrode within an area having a vertical projection apart from the second active region and the second channel.

[0035] In some embodiments, the gate line **201** is a continuous line so that the first gate electrode is structurally connected to the second gate electrode. For example, as shown in FIG. **4**, the gate electrodes as described above are formed along the gate line **201** and the first gate electrode and the second gate electrode are thus structurally connected. In some embodiments, the parallel lines on both the right side and the left side of the gate line **201** are dummy gate lines without any gate electrode formed thereon.

[0036] In some embodiments, the integrated circuit structure shown in FIG. **4** is a CMOS transistor including a pair of a PMOS transistor and an NMOS transistor that are electrically connected to the connection line **401** and **401'** by the two conductive vias **204** and **204'** connecting to each of the gate electrodes of the PMOS transistor and the NMOS transistor, respectively. Referring to FIGS. **6**A and **6**B, FIG. **6**A is a partial and simplified enlargement of the integrated circuit structure shown in FIGS. **4** and **5**, and FIG. **6**B is a cross-sectional view of the structure shown in FIG. **6**A along the AA' line. Specifically, the connection line **401** and **401'**

in FIG. **4** is not shown in FIGS. **5**A and **5**B for the case of making comparison with the integrated circuit structure of FIG. **2**. Compared to the integrated circuit structure shown in FIG. **2**, the integrated circuit structure including two transistors connected to the connection line **401** and **401'** with two conductive vias exhibits reduced RC delay by decreasing half of the gate resistance. Furthermore, due to the structural design, the sheet resistance of the gate line can be reduced to a value close to 0 in the case when the conductive vias **204** and **204'** are disposed on the gate line **201** within the area of the surface of the gate electrodes.

[0037] In some embodiments, the first gate electrode is structurally separated from the second gate electrode. Referring to FIG. **7**, similar to the integrated circuit structure of FIG. **4**, FIG. **7** also shows exemplary plan view of an integrated circuit structure having two transistors with two conductive vias electrically connecting to the two transistors, except that the first gate electrode and the second gate electrode are separated. In some embodiments, the first gate electrode and the second gate electrode are formed on the same gate line **201**, but the connection between the first and second gate electrode is cut by exposing and etching a pattern to the gate line **201** within the space between the first and second gate electrode and filling in a non-conductive material, such as dielectric material, to both physically and electrically separate the first and second gate electrodes.

[0038] In some embodiments, the integrated circuit structure shown in FIG. **4** is a CMOS transistor including a pair of a PMOS transistor and an NMOS transistor that are electrically connected to the connection line **401** and **401'** by the two conductive vias ("via contacts") **101** and **101'** connecting to each of the gate electrodes of the PMOS transistor and the NMOS transistor, respectively.

[0039] Though there is not any specific restriction to the extending direction of the gate line **201**, in some embodiments, the first active region **202** and the second active region **203** are substantially arranged in parallel, and extending in a first direction. In addition, in some embodiments, the first gate electrode is substantially in line with the second gate electrode in a second direction. In some embodiments, the first direction is orthogonal to the second direction.

[0040] In some embodiments, the connection line includes a pair of first connection wires **401** and **401'** disposed on and electrically connecting the first conductive via **204** and the second conductive via **204'**, respectively; a pair of third conductive vias disposed on and electrically connected to the pair of first connection wires **401** and **401'**, respectively; and a second connection wire disposed on and electrically connected to the first connection wires **401** and **401'** through the pair of third conductive vias.

[0041] Thought there is not any specific restriction to the extending direction of the pair of first connection wires, in some embodiments of the present disclosure, the pair of the first connection wires are arranged substantially in parallel, and extending along the first direction. In addition, in some embodiments, the second connection wire extends substantially along the second direction. In some embodiments, the first direction is orthogonal to the second direction.

[0042] In some embodiments, the integrated circuit structure shown in FIGS. **4** and **5** includes a PMOS transistor, an NMOS transistor and a connection line. The PMOS transistor includes a first active region **202** extending in a first direction; a first gate electrode **101** positioned on the first active region **202** and extending in a second direction different from the first direction; a first channel (not shown) in the first active region **202** and under the first gate electrode **101**; and a first conductive via **204** disposed on the first gate electrode **101** and electrically connected to the first gate electrode **101**. Similarly, the NMOS transistor includes a second active region **203** extending in the first direction and parallel to the first active region **202**; a second gate electrode **101'** positioned on the second active region **203** extending in the second direction; a second channel in the second active region **203** and under the second gate electrode **101'**; and a second conductive via **204'** disposed on the second gate electrode **101'** and electrically connected to the second gate electrode **101'**; and a connection line **401** and **401'** electrically connecting the first conductive via **204** and the second conductive via **204'**.

[0043] In some embodiments, the first gate electrode **101** and the second gate electrode **101'** include tantalum gate electrodes, tungsten gate electrodes, tantalum nitride gate electrodes, titanium nitride gate electrodes or polycrystalline silicon gate electrodes. In some embodiments, the first gate electrode **101** and the second gate electrode **101'** include polycrystalline silicon gate electrodes.

[0044] Referring to FIGS. **8**A and **8**B, FIG. **8**A shows exemplary plan view of an integrated circuit structure in accordance with some embodiments of the present disclosure, and FIG. **8**B shows the cross-sectional view of the integrated circuit structure of FIG. **8**A. In FIGS. **8**A and **8**B, the integrated circuit structure further includes a first metal-over-poly layer **801** positioned between the first gate electrode **101** and the first conductive via **204** to electrically connect the first gate electrode **101** and the first conductive via **204**; and a second metal-over-poly layer **801'** positioned between the second gate electrode **101'** and the second conductive via **204'** to electrically connect the second gate electrode **101'** and the second the second conductive via **204'**.

[0045] In some embodiments, the first gate electrode **101** and the second electrode **101'** are formed on the same gate line **201** along the second direction so that the first gate electrode and the second gate electrode **101'** are connected. In some embodiments, the first gate electrode and the second electrode are separated from each other.

[0046] In some embodiments, referring back to FIG. **4** the first conductive via **204** is positioned on the first gate electrode **101** within an area overlapping the first active region **202**, and the second conductive via **204'** is positioned on the second gate electrode **101'** within an area overlapping the second active region **203**. In some embodiments, referring back to FIG. **8**, the first conductive via **204** is positioned on the first gate electrode **101** within an area having a vertical projection apart from the first active region **202**, and the second conductive via **204'** is positioned on the second gate electrode **101'** within an area having a vertical projection apart from the second active region **203** as shown in FIG. **8**.

[0047] Referring to FIG. **9**, FIG. **9** is a flow chart of a method **900** for manufacturing an integrated circuit structure in accordance with some embodiments of the present disclosure. At operation **901**, a first active region and a second active region are formed in a semiconductor substrate. At operation **902**, a gate line including a first gate electrode and a second gate electrode are formed above the first active region and the second active regions to partially overlap the first active region and the second active region. At operation

903, a first conductive via is formed on the first gate electrode and a second conductive via is formed on the second gate electrode. At operation **904**, a connection line is formed above the first conductive via and the second conductive via to electrically connect the first conductive via and the second conductive via.

[0048] In some embodiments, in need for cutting the electrical connection between the gate electrodes of the pair of the transistors in the aforementioned integrated circuit structure manufactured according to the process illustrated in FIG. **9**, the process may further includes cutting the gate line to structurally separate the first gate electrode and the second gate electrode by exposing and etching a pattern to the gate line within the space between the first and second gate electrode and filling in a non-conductive material, such as dielectric material, to both physically and electrically separate the first and second gate electrodes.

[0049] In some embodiments, an integrated circuit structure is provided. In some embodiments, the integrated circuit structure includes a first transistor, a second transistor, a first via and a second via. The first transistor includes a first active region and a first gate electrode overlapping the first active region. The second transistor includes a second active region and a second gate electrode overlapping the second active region. The first via is disposed over the first transistor and the first gate electrode. The second via is disposed over the second transistor and the second gate electrode. The first gate electrode and the second gate electrode are arranged mirror-symmetrically with respect to a symmetry plane.

[0050] In some embodiments, an integrated circuit structure is provided. In some embodiments, the integrated circuit structure includes a first transistor, a second transistor, a first via and a second via. In some embodiments, the first transistor includes a first active region, and a first gate overlapping the first active region. In some embodiments, the second transistor includes a second active region, and a second gate overlapping the second active region. In some embodiments, the first via is over the first active region and the first gate, and being electrically connected to the first gate. In some embodiments, the second via is over the second active region and the second gate, and being electrically connected to the second gate. In some embodiments, the first gate and the second gate are arranged mirror-symmetrically with respect to a symmetry plane.

[0051] In some embodiments, a method for manufacturing an integrated circuit structure is provided. In some embodiments, the method includes forming a first active region and a second active region in a semiconductor substrate, forming a gate line comprising a first gate line portion overlapping the first active region and a second gate line portion overlapping the second active region, forming more than one vias on the gate line, and forming more than one connection lines on the more than one vias respectively.

[0052] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other operations and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

[0053] Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure of the present invention, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present invention. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

What is claimed is:

1. An integrated circuit structure, comprising:

a first transistor, including a first active region and a first gate electrode overlapping the first active region;

a second transistor, including a second active region and a second gate electrode overlapping the second active region;

a first via disposed over the first transistor and the first gate electrode; and

a second via disposed over the second transistor and the second gate electrode, wherein the first gate electrode and the second gate electrode are arranged mirror-symmetrically with respect to a symmetry plane.

2. The integrated circuit structure according to claim **1**, wherein the first gate electrode and the second gate electrode comprise polycrystalline silicon.

3. The integrated circuit structure according to claim **1**, wherein

the first transistor further includes a first source region and a first drain region in the first active region, wherein a first channel region is defined between the first source region and the first drain region, and the first gate electrode overlaps the first channel region; and

the second transistor further includes a second source region and a second drain region in the second active region, wherein a second channel region is defined between the second source region and the second drain region, and the second gate electrode overlaps the second channel region.

4. The integrated circuit structure according to claim **1**, further comprising:

a first metal-over-poly layer over the first gate electrode and the first via; and

a second metal-over-poly layer over the second gate electrode and the second via.

5. The integrated circuit structure according to claim **1**, wherein the first via and the second via are arranged mirror-symmetrically with respect to the symmetry plane.

6. The integrated circuit structure according to claim **1**, wherein

the first via is positioned over the first gate electrode within an area overlapping the first active region, and

the second via is positioned on the second gate electrode within an area overlapping the second active region.

7. The integrated circuit structure according to claim **1**, further comprising:

an isolation, disposed between the first active region and the second active region, wherein a vertical projection of the first via overlaps the isolation and is proximal to the first active region, and a vertical projection of the second via overlaps the isolation and is proximal to the second active region.

8. The integrated circuit structure according to claim **1**, further comprising:

a first dummy gate over the first active region and the second active region, and being next to a first side of the first gate electrode and a first side of the second gate electrode.

9. The integrated circuit structure according to claim **8**, further comprising:

a second dummy gate over the first active region and the second active region, and being next to a second side of the first gate electrode and a second side of the second gate electrode, the second side of the first gate electrode being opposite from the first side of the first gate electrode, and the second side of the second gate electrode being opposite from the first side of the second gate electrode.

10. A method for manufacturing an integrated circuit structure, comprising:

forming a first active region and a second active region in a semiconductor substrate;

forming a gate line comprising a first gate line portion overlapping the first active region and a second gate line portion overlapping the second active region;

forming more than one vias on the gate line; and

forming more than one connection lines on the more than one vias respectively.

11. The method according to claim **10**, further comprising cutting the gate line to structurally separate the first gate line portion and the second gate line portion.

12. An integrated circuit structure, comprising:

a first transistor comprising:

a first active region; and

a first gate overlapping the first active region;

a second transistor comprising:

a second active region; and

a second gate overlapping the second active region;

a first via over the first active region and the first gate, and being electrically connected to the first gate; and

a second via over the second active region and the second gate, and being electrically connected to the second gate,

wherein the first gate and the second gate are arranged mirror-symmetrically with respect to a symmetry plane.

13. The integrated circuit structure according to claim **12**, wherein each of the first gate and the second gate comprise tantalum, tungsten, tantalum nitride, titanium nitride or polycrystalline silicon.

14. The integrated circuit structure according to claim **12**, wherein

the first active region comprises: a first source region and a first drain region, wherein a first channel region is between the first source region and the first drain region, and the first gate overlaps the first channel region; and

the second active region comprises: a second source region and a second drain region, wherein a second channel region is between the second source region and the second drain region, and the second gate overlaps the second channel region.

15. The integrated circuit structure according to claim **12**, further comprising:

a first conductor electrically coupled to and over the first gate and the first via, the first via being between the first conductor and the first gate; and

a second conductor electrically coupled to and over the second gate and the second via, the second via being between the second conductor and the second gate.

16. The integrated circuit structure according to claim **12**, wherein the first via and the second via are arranged mirror-symmetrically with respect to the symmetry plane.

17. The integrated circuit structure according to claim **12**, wherein

the first via is over the first gate within a first area overlapping the first active region, and

the second via is on the second gate within a second area overlapping the second active region.

18. The integrated circuit structure according to claim **12**, further comprising:

an isolation region between the first active region and the second active region, wherein a first vertical projection of the first via overlaps the isolation region and is proximal to the first active region, and a second vertical projection of the second via overlaps the isolation region and is proximal to the second active region.

19. The integrated circuit structure according to claim **12**, further comprising:

a first dummy gate over the first active region and the second active region, and being next to a first side of the first gate and a first side of the second gate.

20. The integrated circuit structure according to claim **19**, further comprising:

a second dummy gate over the first active region and the second active region, and being next to a second side of the first gate and a second side of the second gate, the second side of the first gate being opposite from the first side of the first gate, and the second side of the second gate being opposite from the first side of the second gate.

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